

Advanced Modules — Optional Follow-On Topics

Foundations of Clinical Research: From Clinical Question to Publication

Organisation: Neudata — #ClearDataClearImpact

> **Outline only** — module materials will be developed on demand, after the 8-session course has run.

> The 8-session course takes participants from a clinical question to a draft manuscript, with the analyses run by instructors. These optional modules are for smaller groups who want to go further: doing the analyses themselves, or learning a specialised method that the course only introduced.

1. How the modules work

Who	Graduates of the 8-session course, 6–12 per module. Each module can run on its own; they need not be taken in order.
Format	1–2 sessions of 3 hours each (the same rhythm as the course), plus 3–6 hours of homework on the participant's own project or a course dataset.
Software	R and RStudio on participants' own laptops (free). The hands-on modules assume the free Neudata course <i>Clinical Data Analysis in R</i> has been completed.
Data	FLUID-ICU (the course dataset) for practice, then the participant's own data where ethics approval and data-protection rules allow. Purpose-built teaching datasets for the meta-analysis and diagnostic-accuracy modules.
Output	Every module ends with a concrete product: a reproducible script, an analysis section, a forest plot, a model report or a checklist-based appraisal.
Link to the course	The course <i>introduced</i> each topic (shown as "Course link" below); the module turns it into hands-on practice.

2. Module A — Hands-on R analysis of your own data

Course link: Sessions 5–6 (data cleaning, Table 1, tests, regression, Kaplan–Meier/Cox, missing data — all demonstrated by the instructor) and the FLUID-ICU R scripts.

Aims. Participants will be able to (1) import, clean and document a real dataset reproducibly, with a cleaning log; (2) produce Table 1 and publication-quality figures; (3) run and check the analyses in a pre-written statistical analysis plan (tests, linear/logistic regression, Kaplan–Meier and Cox, a multiple-imputation sensitivity analysis); and (4) export results into the manuscript template.

Prerequisites. Completion of the 8-session course; *Clinical Data Analysis in R* completed; a laptop with R and RStudio; ideally a dataset of one's own (or a proposal from the course that has collected

data) with ethics approval and a written SAP.

Outline (2 sessions).

Session	Content	Product
A1 — From raw data to Table 1	Reproduce the FLUID-ICU cleaning script line by line; adapt it to one's own data (duplicates, sentinel values, implausible values, derived variables); cleaning log; Table 1 and figures; version control and backups	Clean dataset + cleaning log + Table 1 for one's own data
A2 — From SAP to Results	Run the SAP: unadjusted comparisons, adjusted model (logistic, linear or Cox), checking assumptions, multiple imputation as sensitivity analysis; tables and forest plot; writing the statistical-methods paragraph	Results section draft + reproducible script

3. Module B — Systematic review and meta-analysis

Course link: Sessions 3–4 (systematic reviews, PRISMA flow, forest plots, heterogeneity, risk-of-bias tools) and Session 2 (searching).

Aims. Participants will be able to (1) write a systematic-review protocol and register it in PROSPERO; (2) build a reproducible multi-database search and screen records in pairs; (3) extract data and assess risk of bias (RoB 2 or ROBINS-I); and (4) pool effect estimates (fixed vs random effects), interpret heterogeneity (I^2) and draw a forest and funnel plot, reported with PRISMA 2020.

Prerequisites. Completion of the 8-session course; for the analysis part, *Clinical Data Analysis in R* (the meta-analysis is run with a ready-made script); a review question (PICO) agreed with the module lead.

Outline (2 sessions).

Session	Content	Product
B1 — Protocol, search and screening	Review question and eligibility; PROSPERO; search strategy across databases; screening in pairs; PRISMA flow diagram; data-extraction form	Registered (or ready-to-register) protocol + search strategy
B2 — Bias, pooling and reporting	Risk-of-bias assessment; pooling ORs/RRs in R; fixed vs random effects; I^2 and subgroup ideas; funnel plot and small-study effects; certainty of evidence (GRADE, introduced); PRISMA 2020	Forest plot + draft Results for a small teaching meta-analysis

4. Module C — Prediction models and machine learning

Course link: Session 6 (overfitting, prediction vs explanation, calibration and discrimination, TRIPOD+AI introduced).

Aims. Participants will be able to (1) tell a prediction question from a causal question; (2) estimate the sample size needed to develop a model and avoid overfitting; (3) develop a simple logistic prediction model and assess discrimination (AUC) and calibration, with internal validation (bootstrapping); (4) understand what machine-learning methods add and where they mislead; and (5) appraise a published model with TRIPOD+AI and PROBAST.

Prerequisites. Completion of the 8-session course; *Clinical Data Analysis in R*; comfortable with logistic regression output.

Outline (2 sessions).

Session	Content	Product
C1 — Building a model properly	Prediction vs causation; candidate predictors; sample size for prediction models; developing a logistic model on FLUID-ICU (28-day mortality); discrimination and calibration plots; bootstrapped internal validation	Model report with AUC, calibration plot and optimism-corrected performance
C2 — Machine learning and appraisal	Penalised regression and a tree-based method compared with logistic regression; data leakage and other pitfalls; external validation; fairness; reading a published AI/ML model with TRIPOD+AI and PROBAST	Structured appraisal of one published prediction or ML model

5. Module D — Diagnostic-accuracy studies

Course link: Session 4 (sensitivity, specificity, predictive values and prevalence, ROC curves, likelihood ratios, QUADAS-2 introduced).

Aims. Participants will be able to (1) design a diagnostic-accuracy study (index test, reference standard, spectrum of patients, blinding); (2) calculate sensitivity, specificity, predictive values and likelihood ratios with 95% CIs, and draw a ROC curve with its AUC; (3) choose a cut-off sensibly; and (4) report with STARD 2015 and appraise with QUADAS-2.

Prerequisites. Completion of the 8-session course. R is used only in the second half (a ready-made script); participants without R can follow with a spreadsheet calculator.

Outline (1 session, optional second session for a project).

Session	Content	Product
D1 — Design, analysis and reporting	Diagnostic questions and designs; common biases (spectrum, verification, incorporation); 2×2 accuracy table with CIs; predictive values across prevalences; ROC/AUC and cut-offs in R (e.g. lactate for 28-day mortality in FLUID-ICU); STARD 2015 and QUADAS-2	Completed accuracy analysis + STARD-based outline of a study

Session	Content	Product
D2 (optional) — Your own diagnostic question	Protocol for a unit-level diagnostic-accuracy study, sample size for sensitivity/specificity, data-collection form	Protocol outline

6. Module E — Advanced survival and longitudinal analysis

Course link: Session 6 (censoring, Kaplan–Meier, log-rank, Cox HR; demonstrated on FLUID-ICU Question 1).

Aims. Participants will be able to (1) fit and check Cox models (proportional-hazards assumption, time-varying effects); (2) handle competing risks (e.g. ICU discharge alive vs death) and immortal-time bias; (3) analyse repeated measurements over time (e.g. daily fluid balance or SOFA) with mixed-effects models; and (4) report these analyses clearly.

Prerequisites. Completion of the 8-session course; *Clinical Data Analysis in R*; Module A recommended.

Outline (2 sessions).

Session	Content	Product
E1 — Beyond the basic Cox model	Checking proportional hazards; time-varying covariates; immortal-time bias; competing risks (cumulative incidence, Fine–Gray, introduced); reporting time-to-event results	Survival analysis of FLUID-ICU with PH checks and a competing-risk comparison
E2 — Longitudinal data	Long vs wide data; plotting trajectories; linear mixed-effects models for daily measurements; missing data in repeated measures; when to ask a statistician	Mixed-model analysis of daily fluid balance with a short Results paragraph

7. Choosing a module

If you want to...	Take
Analyse the data from your own project yourself	A (then E if your outcome is time-to-event or repeated)
Summarise all the evidence on a question	B
Build or judge a risk score, prediction tool or AI model	C
Test how good a test, score or sign is at detecting a condition	D
Go deeper on time-to-event or repeated-measures outcomes	E

Key resources for each module are listed in [Course/References/further_reading.md](#) (Cochrane Handbook and PRISMA 2020 for B; TRIPOD+AI and the Riley sample-size paper for C; STARD 2015 and QUADAS-2 for D; *Statistics at Square One* and the Bland–Altman Statistics Notes for A and E).